PROJECT DSA: Shortest City Map Path Finder

Narrative Video Title: *<LOOK BEFORE YOU LEAVE>*

PLOT: *The story begins with C1, a high school student, rushing to prepare for school. Despite having promised to wake up early, C1 ends up running late. While chatting with his friend (C3) on the phone, he confidently claims he will reach school in 15 minutes. Little does he know, his overconfidence will soon be tested by real-world traffic challenges.*

INTRO:

[voice of C1/offscreen]: We've all experienced the frustration of navigating through a bustling city. Congested streets, unexpected delays, and confusing roadways can make getting to your destination a real challenge. But have you ever wondered why congestion happens? And how can we navigate through it more efficiently?

>>> -added effects (a clip of busy bustling cities)

SCENE 1: The Student

... Thumping. Thumping… (run there, there and there)

>>C1 (nervously check on his phone):

Naku! po malalate na naman. (message chat with his friend while busy in getting ready for school)

(A little while…. C1 phone ring…)

>>C3(angrily speaking): hello pre!, sabi mo maaga kang matutulog para maaga magising. Ehhh!... Tapos nagchat kang papunta palang.

>>C1(excitedly speaking): Oo na pre!.Ikaw na malapit bahay sa school!.

>>C1(hopefully speaking): Cge na kita kits nalang mamaya. Basta… papunta nako.

>>C3(worriedly speaking): Cge cge.. Basta siguraduhin mo makakaabot ka ahh! May kwiz pa naman tayo bago mag lesson sa DSA

>>C1(excitedly speaking): Oo pre! Mga… ahmm.. 15 minutes andiyan nako.

>>C2(hopefully speaking): Cgecge bye na. ingat pre bye!

>>C1(pans the camera on C1 charging phone)

>>C1(annoyed face and irritatedly speaking): Naku po naman talaga! di naman pala nakasaksak nang maayos. late na nga tayo! Lowbat pa ang phone.

(C1 hurriedly left home going to school).

SCENE 2: Waiting Friends in School (*changing the setting to school setting C3 and C4*)

(*-Camera Pans to C3 and C4*)

(*added effects (a clip of the schoo*)

>>C4(excitedly walking): Pre! Ang aga mo talaga sanaol na lang malapit ang bahay.

>>C4(suspicious looked): Wait lang saan na naman si C1(name) …

>>C3(worriedly talking): Ano pa ba pre!, Wala pa… hay.. kala ko pa naman magiging mas maaga na siya dahil natulog nang maaga.

C4(worriedly speaking): haluhh pano yan diba may kwiz pa tayo sa DSA. Diba unang subject natin un.

Anong oras kaya yun(*swaying gesture of head*) makakarating na naman…

C3(excitedly speaking): kaya nga haystt tinawagan ko nga… at ang sabi on the way na daw pero 15 mins pa daw.

C4: ahhaahha! Grabe naman yan kung andito lang yun…. magaaway na naman kayo. Hayaan muna. Sana! Na-iapply niya yung topic natin sa DSA yung (**Dijkstra's Algorithm**).

C4: (EXPLAINING OFFSCREEN the details of how to optimize city navigation, let's lay the foundation by introducing two key concepts: Graph Theory and Dijkstra's Algorithm.)

SCENE 3: The Ride Begins (*changing the setting (C1 and C2) (Isang abalang kalsada sa lungsod*)

C1(panting speaking): Pre! Ikaw pala yan

(*Shortness of breath* *Panting*)

C2(suprisingly face and speak): Uyy! Musta…bat muka ka atang nagmamadali…Teka malalate ka na naman noh hay nako dikapa din nagbago kahit nung hayskul nalalate kapa din pre.

C1(a lil' bit out of breath speakingly) : But-i nalan-gg na-k-ita ki-ta. Pasab-ay nga Pre--. Sa school lang— plss?

C2(still pondering): (*Thinking face *)

C1(excitedly speaking): Cge na Pre! Minsan lang naman.

C2(Irritatedly speaking): Huh?!! Anong minsan. Ehh! Lagi tayong sabay umalis nung hayskul pa tayo. Dali na baka mas malate kapa!

C1(excitedly rode the vehicle then check his phone)

C1(franticly faced): Luhh! Lowbat na agad… Paano ko malalaman ang pinakamabilis na ruta neto? Kaya kayang makaabot tayo sa loob ng 15(labinlimang) minuto?

C2(Smugly faced): Hehh! Relax. Meron din ako niyan Pero sa nakikita kong trapik dito ngayon mahihirapan tayo.

C2(*sees his phone and the real time ETA is within 22 mins.)

C1: Lagot na talaga sabi before ako umalis 15 mins daw aabot na sa school. Hayyyy!!!.

C2: Kala ko ba nalesson na sainyo yung Graphs to Real World traffic sa DSA! Dimo ba alam na nagbabago ang flow nang trafic depende sa weights or yung mga complex scenario na di inaasahan.

C1: Ahh yun oo na yun nga ung kwiz ngayon ehh!. Kaya ayaw ko malate.Meron din pla kayong ganun sa IT. Nagbabago pala ang estimated time(*tango tango*)

C2: Syempre naman!....Kahit naman pala maleyt ka parang wala ka naman isasagot pre, joke!. So habang nagmamaneho ako ieexplain ko sayo kung bakit.

----OFFSCREEN---

-added clip of REAL busy roads(morning)

(* add a diagrams or demos showcasing the algorithm of the coding group, prob ppt**)**

C2:

The flow of traffic in a city is influenced by a complex web of choices made by individual drivers, each trying to pick the quickest or most convenient route. Similarly, in the world of graph-based algorithms, the pattern of movement across a network of nodes and edges often reflects the same interdependent dynamics.

C1: ohhh parang naalala ko naung lesson!

C2: wehh? Talaga ba? Bago paman makarating tayo at ang lahat let me get into details of how to optimize city navigation, let's lay the foundation by introducing two key concepts: **Graph Theory** and **Dijkstra's Algorithm**.

C1: yup yup natatandaan kona ito ba yung algorithm na (**Dijkstra's Algorithm**)

SCENE 4:--- Relating Graphs to Real-World Traffic --- (*OFFSCREEN*)

C4:

Imagine a city map as a **graph**, where: (***Clip of a city map and (PPT of a table representing this terms***.)

- **Nodes** represent key locations such as intersections, landmarks, or addresses.
- **Edges** represent roads connecting these locations.
- **Weights** represent the travel cost on each road, which could be distance, travel time, or even toll fees

C2:

When navigating this graph, the challenge is finding the **shortest path** from a starting node (your current location) to a destination node (your goal). But just like in a real-world city, delays (like traffic congestion) influence the weights of edges, making the shortest path calculation dynamic and sometimes non-intuitive.

(*Pans the Camera to C1*)

C1: so kaya pla nagupdate din yung oras nung estimated time ko sa oras na kailangan para makarating sa school

C2: Yep!

This is where (**Algorithm used by the coding group)** comes into play.

(***clip of a bustling city but the speed is 2x***)

(***BACK TO OFFSCREEN)

C2：(**Dijkstra's Algorithm**) ensures that you always find the most optimal route based on the current weights (delays or distances). It's like having an intelligent navigation app that adapts to real-time conditions.

(*Pans the Camera to C1*)

C1: so eto yun sinasabi mong tinatake into account nang algorithm to represent both the distance of the location and pati narin yung instances delays while palapit si user sa pupuntahan… wow amazing.

(***BACK TO OFFSCREEN)

C2: di lang yan dahil lang sa may congested traffic or weights instance is magbabago na agad, thus

This process not only optimizes your route but can also help manage city-wide traffic patterns by identifying bottlenecks or providing alternative paths Thus, (Algorithm used by the coding) functions as an intelligent navigation tool that can adjust to varying conditions in real-time.

C1: so kung iisipin if iaaply sa sitwasyon ko ngayon edi...

--- OFFSCREEN ----

Let’s put this into perspective:

---(Animated Clip of a city map of location A and location B)---

- Imagine you’re in a city, and you need to get from Point A (Home) to Point B (School).
- The city’s intersections and roads are mapped as a graph.

(**Dijkstra's Algorithm**)will compute the **shortest path** based on travel time (weights) while considering congestion on different roads (variable edge weights).

SCENE 5: Arriving Late.

(---changing the setting to school setting (C3 and C4)(while reviewing before the class---)

C4 (deeply thinking): Saan na kaya pre si (C1) sabi niya sau 15 mins lang siya, pero mag 15 mins na ohh 4 mins nalang, may quiz pa naman kanino nako neto mangagaya *hshshs* joke (“imitate a joke hand figure”).

C3(worriedly speaking): Ewan ko ba dun maaga na nga natapos nung naka assign sa kanya at maaga nading natulog pero late padin…

C4: Ireview nga kita! Para Makita ko kung nagreview k aba talaga

C3: Cge ba! Kala mo naman si (C1) lang maasahan natin sa gantong Gawain sating tatlo

C4: Cge nga—teka pili ko nang mahirap hirap—ahh heto!

Why does this Short-Map City Finder Matter?

C3: Sus ganto kasi yan:

----OFFSCREEN----

C3: (voiceover)

(***PPT about why map pathfinders is important to our real world.***)

-Efficient pathfinding isn’t just about personal convenience; it’s about optimizing larger systems:

- Emergency Services: Rapid access for ambulances and fire trucks is crucial for saving lives and ensuring safety.

- Public Transportation: Streamlined routes can drastically cut down on delays, promoting more effective transit.

- Urban Planning: Smart design and infrastructure can create more functional and organized cities, enhancing the quality of life for residents.

C4: Wow! Naman nagreview talaga siya ohh! Parang kumain ka nang libro.

C3: Cge ikaw naman nga tanungin ko sa tingin moba sa mundong ito importante ba ang mga gantong concept na alamin.

C4: luhh Andaya bat parang magpapasagot ba si maam nang paragraph style sa kwiz. Ang masasabi ko lng diyan is.

--- OFFSCREEN----

(***PPT representing real-life applied importance of pathfinders in our real world with the evolving cities.***)

C4(voiceover):

Understanding concepts like graph theory and Dijkstra's Algorithm not only helps solve navigation problems but also builds a foundation for addressing more complex systems. By exploring these principles, you can better appreciate the interplay between algorithms and real-world challenges.

Always remember: As cities evolve and networks become more interconnected, tools like the City Map Shortest Path Finder aren't just innovations—they're necessities for smarter living.

SCENE 6: THE LESSON- REALITY OF CONEGESTION (*Setting: School gate, C1and C2 arrives at the school and will exit the vehicle.*)

C1: Oyy salamat naman nakarating din sa school. Hatid muna ko pre diretso sa building natin dun Karin naman papunta ehh…

C2: Late kana din pre. Tingnan mo ohh.. 22 minutes na before tayo umalis. Sana naman pre may natutunan ka sa algorithm nung lesson niyo di lang basta

basta dahil 15 mins ang papunta sa school at bahay ayy yun na lang lagi.

C1 and C2(arrive at the department of CCS bldg..)

C1: cge na pre lamats nang madami sana magamit ko tong natutuhan ko sa quiz pagdating ko sa room.

C2(waves from the parking-vehicle)

C1(running to the classroom)

C3 & C4 (still reviewing) (***camera pans to c3 and c4)

C4: Eyy! Nakaabot Kadin --- kahit late ka naman talaga lagi.

C3: Sabi mo Pre! 15 mins lampas 20 mins na ohh.

C1(lil' bit breathing heavily): Wala pa--- ba si Maam?

C4: Wala pa naman late din siya ehh, buti nalang talaga.

C1: Hayystt kala ko malalate nako. Ang alam ko kc dati pa 15 mins ung estimated time papunta dito ehh.Nagbabago play un buti nalang nadatnan ko si (C2) at nakisakay ako

C4: Sa susunod iaapply mo na ung konsepto nung Graph theory and (algorithm used by the coding group)

C1: Wow! Parang nag review siya ahh!

C4: Syempre naman kalako malalate ka ehh edi walang makokopyahan AHAHHAH!.

C3: Kaya dapat Hindi lang ito tungkol sa pag-alam ng mga kalsada,kasi di natin alam may instances na bigla nalang may congestions kahit pa yan weekend or weekdays.

C1: Oo na!

C4: Guyss andiyan na daw si maam Tara na Pre!. Buti nalang talaga nakaabot ka.

Scene 7: (ROLL CREDITSS< THANKS EVERYONE)